

```
cd sumo-0.22.0/
./configure --with-fox-includes=/usr/include/fox-
1.6 --with-gdal-includes=/usr/include/gdal --with-
proj-libraries=/usr --with-gdal-libraries=/usr
--with-proj-gdal
make
sudo make install
```

- To start SUMO from the command line, run the following command:

```
sumo or sumo-gui
```

OpenStreetMap is a free editable map of the whole world. It creates and provides free geographic data such as street maps, to anyone who wants them. The project was started because most maps that you think of as 'free' actually have legal or technical restrictions on their use, preventing people from using them in creative, productive or unexpected ways.

You can use <http://www.openstreetmap.org>.

`org#map=5/51.500/-0.100` to download your map.

An OpenStreetMap file contains a map with the following features related to traffic simulation:

- The nodes and their connections define the position and form of all streets and junctions.
- The type of a street specifies its size and importance (key highway).
- The speed limit of a street is usually determined implicitly by law. The implicit value may depend on the value of the highway attribute. In case the speed limit differs from the implicit legal value, it is given explicitly through the key maximum speed of a street.

- The total number of lanes (for both directions) has an implicit value depending on the highway property. If the real value differs from the default value, it can be given through the key lanes.
- The position of every traffic light is described as a node with the key-value pair `highway=traffic_signals`.
- One-way streets are marked with the key-value pair `oneway=yes`.

Figure 1 shows the map that I downloaded from openstreetmap.org.

You need to select the map that you want to simulate and export it. After you export the map, it will ask you to download it. Save the file. Copy the downloaded file to your working directory.

Now run the following commands, one by one.

Netconvert imports digital road networks from different sources and generates road networks that can be used by other tools from the package. It imports the road network stored in `map.osm` and stores the SUMO-network generated from this data into `map.net.xml`:

```
netconvert --osm-files map.osm -o map.net.xml
```

OSM-data not only contains the road network but also a wide range of additional polygons such as buildings and rivers. These polygons can be imported using *polyconvert* and then added to a *sumo-gui-configuration*.

To interpret the OSM-data, an additional *typemap* file is required.

Go to <http://sumo.de/wiki/Networks/Import/> *OpenStreetMap*, copy the *typemap* file and save it to your working directory with the name *typemap.xml*.

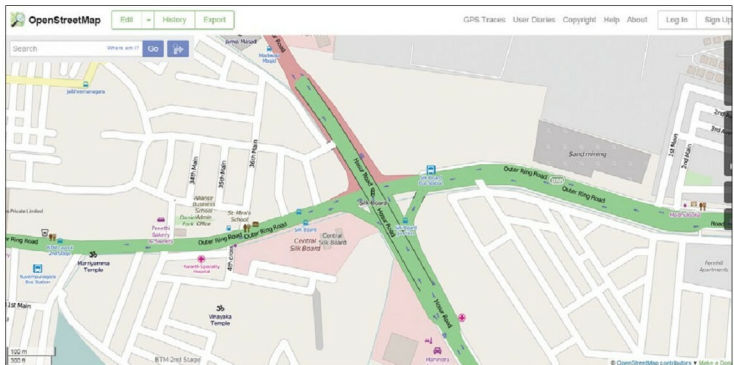


Figure 1: Map exported from www.openstreetmap.org